Accepted Binder Models — Binding Interaction Analysis

BindCraft A$ $42 Campaign (9CO4 C/E/G)

2026-05-08

1. Summary of All 7 Accepted Designs

Rank	Design	Len	MW	i_pTM	pTM	pLDDT	i_pAE	B_pLDDT	B_RMSD	S.Hyd	SC	dG	dSASA	nIR	nIHb	nUHb	H.RMSD	R.Cl	K	M	HS	Helix%	Notes
1	s120913_mp1	79	8.7	0.81	0.84	0.88	0.19	0.90	2.61	0.26	0.68	-93.7	2929	32.5	7.5	1.5	1.02	0	1.5	1.0	14/18	67.1	Clean
2	s120913_mp2	79	8.7	0.81	0.84	0.88	0.19	0.89	2.18	0.28	0.69	-98.9	3146	39.5	11.5	1.0	1.67	0.5	1.0	2.0	14/18	65.8	R.Cl
3	s716952_mp20	87	9.6	0.76	0.80	0.89	0.21	0.89	1.47	0.34	0.66	-74.7	2541	29.5	4.0	2.0	3.09	0	1.0	3.0	13/18	77.0	Clean, NEW
4	s480128_mp13	82	9.0	0.72	0.80	0.89	0.20	0.86	2.15	0.34	0.71	-73.7	2042	24.0	3.5	2.0	1.15	0.5	1.0	0.0	11/18	85.4	R.Cl
5	s480128_mp17	82	9.0	0.76	0.82	0.92	0.18	0.95	0.90	0.34	0.68	-71.0	2141	25.5	3.5	2.5	1.51	0.5	1.0	0.0	11/18	85.4	R.Cl
6	s967366_mp11	82	9.0	0.79	0.84	0.94	0.17	0.95	1.11	0.32	0.76	-71.8	2003	21.5	3.5	3.0	1.12	0	1.5	3.0	10/18	82.9	Clean
7	s311665_mp6	68	7.5	0.73	0.80	0.89	0.20	0.88	1.22	0.30	0.67	-62.7	2029	25.0	3.5	3.5	2.50	0	0.5	3.0	9/18	82.4	Clean

Column key: Len = binder length (aa), MW = molecular weight (kDa), SC = shape complementarity, dG = binding energy (REU), dSASA = buried surface area (Å^2), nIR = interface residues, nIHb = interface H-bonds, nUHb = unsatisfied H-bonds, H.RMSD = hotspot RMSD (Å), R.Cl = relaxed clashes, K/M = interface Lys/Met count, HS = hotspot contacts.

2. Binding Interaction Analysis — Overview

Design	HS	Chain C	Chain E	Chain G	Binder IF Res	H-bond pairs	Hydrophobic pairs	Salt bridges	Dominant type
s120913_mp1	14/18	6/6	5/6	3/6	42	12	191	4	Mixed (H-bond + Hydro)
s120913_mp2	14/18	6/6	5/6	3/6	45	13	203	2	Mixed (H-bond + Hydro)
s716952_mp20	13/18	3/6	4/6	6/6	33	4	163	0	Hydrophobic
s480128_mp13	11/18	3/6	4/6	4/6	32	5	150	0	Hydrophobic
s480128_mp17	11/18	3/6	4/6	4/6	31	5	189	0	Hydrophobic
s967366_mp11	10/18	2/6	3/6	5/6	25	4	163	0	Hydrophobic
s311665_mp6	9/18	3/6	3/6	3/6	27	5	142	0	Hydrophobic

2.1 Hotspot Residue Contact Map

Which hotspot residues are contacted ($<$5 Å) by each design:

	Chain C						Chain E						Chain G					
Design	Y10	E11	H13	H14	Q15	K16	Y10	E11	H13	H14	Q15	K16	Y10	E11	H13	H14	Q15	K16
s120913_mp1	•	•	•	•	•	•	•	•		•	•	•	•				•	•
s120913_mp2	•	•	•	•	•	•	•	•		•	•	•	•				•	•
s716952_mp20	•				•	•	•	•			•	•	•	•	•	•	•	•
s480128_mp13	•				•	•	•	•			•	•	•	•			•	•
s480128_mp17	•				•	•	•	•			•	•	•	•			•	•
s967366_mp11					•	•	•				•	•	•	•	•		•	•
s311665_mp6	•				•	•	•				•	•	•				•	•

Observations:

- Q15 and K16 are contacted by all 7 designs on all 3 chains — most accessible hotspots.
- Y10 is contacted by 6/7 designs on at least chain C.
- H13/H14 are the hardest to reach — only s120913 (chain C) and s716952 (chain G) contact them.
- s120913 is the only scaffold achieving 6/6 contacts on any single chain (chain C).

3. Per-Design Hotspot Interaction Tables

3.1 Rank 1: s120913_mpnn1 (79 aa, 14/18 hotspots, 4 salt bridges)

Hotspot	Binder Res	AA	Dist (Å)	Interaction Type
C:Y10	B:3, B:4, B:7	T, R, L	2.1	Hydrophobic
C:E11	B:4, B:7	R, L	3.3	vdW
C:H13	B:78, B:79	E, E	2.6	Hydrophobic
C:H14	B:77, B:78, B:79	V, E, E	1.8	**H-bond**, Hydrophobic
C:Q15	B:8, B:76, B:77, B:78, B:79	W, I, V, E, E	2.3	Hydrophobic
C:K16	B:8, B:76, B:77, B:78, B:79	W, I, V, E, E	2.0	**H-bond**
E:E11	B:7	L	2.8	Hydrophobic
E:H14	B:79	E	4.4	vdW
E:K16	B:8, B:79	W, E	1.8	**H-bond, Salt bridge**
E:Q15	B:8, B:11, B:12, B:15	W, A, T, L	1.7	**H-bond**, Hydrophobic
E:Y10	B:3, B:7, B:50, B:52	T, L, K, F	2.0	Hydrophobic
G:K16	B:15	L	2.8	Hydrophobic
G:Q15	B:11, B:12, B:15, B:18	A, T, L, R	1.9	**H-bond, Salt bridge**
G:Y10	B:7, B:11, B:14, B:47, B:52	L, A, Q, A, F	1.7	**H-bond**, Hydrophobic

Key binder residues: B:8 W (H-bond + hydro with C/E:Q15, K16), B:18 R (**salt bridge** with G:Q15), B:79 E (**salt bridge** with E:K16), B:14 Q (H-bond with G:Y10), B:77 V (H-bond with C:K16).

Sequence: MDTREQLWWFATAQLLVRHIIEHMRAVGDTSQLARWEADLEILEERARRKEFTIPEDTEIYRLMKTLKENTKGHKIVEE

3.2 Rank 2: s120913_mpnn2 (79 aa, 14/18 hotspots, 2 salt bridges)

Hotspot	Binder Res	AA	Dist (Å)	Interaction Type
C:Y10	B:1, B:2, B:3, B:4, B:7	M, P, T, R, L	2.0	Hydrophobic
C:E11	B:4, B:7	R, L	2.9	vdW
C:H13	B:4, B:78, B:79	R, E, E	3.3	vdW
C:H14	B:77, B:78, B:79	V, E, E	1.9	**H-bond**, Hydrophobic
C:Q15	B:4, B:8, B:76, B:77, B:78	R, W, I, V, E	2.1	**H-bond**, Hydrophobic
C:K16	B:8, B:76, B:77, B:78, B:79	W, I, V, E, E	1.9	**H-bond**
E:E11	B:7	L	2.7	Hydrophobic
E:H14	B:79	E	4.4	vdW
E:K16	B:8, B:15, B:79	W, L, E	3.2	vdW
E:Q15	B:7, B:8, B:11, B:12, B:15	L, W, A, T, L	1.8	**H-bond**, Hydrophobic
E:Y10	B:2, B:7, B:50, B:52	P, L, K, F	2.1	Hydrophobic
G:K16	B:15	L	2.6	Hydrophobic
G:Q15	B:11, B:15, B:18	A, L, R	1.8	**H-bond, Salt bridge**
G:Y10	B:7, B:11, B:14, B:47, B:52	L, A, Q, A, F	1.7	**H-bond**, Hydrophobic

Key binder residues: B:8 W (H-bond with C/E:Q15), B:18 R (**salt bridge** with G:Q15), B:14 Q (H-bond with G:Y10), B:76 I (H-bond with C:Q15), B:77 V (H-bond with C:K16). Same scaffold as Rank 1, different MPNN sequence. Strongest binding energy (dG = -98.9 REU), largest BSA (3146 Å^2).

Sequence: MPTREKLWWFATAQLLVRHIIEHMRARGDTSQLAQWEADLEILEENARKKIFEIPEDTPIYRLMKTLKENTKGHEIVEE

3.3 Rank 3: s716952_mpnn20 (87 aa, 13/18 hotspots — NEW)

Hotspot	Binder Res	AA	Dist (Å)	Interaction Type
C:Y10	B:3	I	3.0	Hydrophobic
C:Q15	B:6, B:8, B:9	T, V, E	1.7	**H-bond**, Hydrophobic
C:K16	B:8	V	3.4	vdW
E:E11	B:58	M	4.5	vdW
E:Y10	B:3, B:4, B:6, B:58	I, P, T, M	2.4	**H-bond**, Hydrophobic
E:Q15	B:6, B:7, B:8	T, I, V	1.7	Hydrophobic
E:K16	B:8	V	3.5	vdW
G:Y10	B:4, B:54, B:55, B:58, B:59	P, V, Y, M, L	2.2	Hydrophobic
G:E11	B:49, B:50, B:54, B:58	D, D, V, M	2.4	Hydrophobic
G:H13	B:48, B:49, B:50	H, D, D	2.3	Hydrophobic
G:H14	B:48, B:49	H, D	2.0	Hydrophobic
G:Q15	B:7, B:43, B:46, B:48	I, H, L, H	2.1	Hydrophobic
G:K16	B:7, B:48	I, H	4.0	vdW

Key binder residues: B:6 T (H-bond with C:Q15), B:4 P (H-bond with E:Y10), B:48 H (contacts G:H13, G:H14, G:Q15, G:K16 — multi-hotspot anchor), B:58 M (contacts E:E11, E:Y10, G:E11, G:Y10 — cross-chain bridge). Only design with 6/6 hotspot contacts on chain G including rare H13/H14.

Sequence: EGIPPTIVESFQFMMEISKVWDAMPEEYRVPLKELFTKLIWLHINLPHDDSEEVYRKMLEAGRLYEEIKKLWEEAMKIPEVRAAAEK

3.4 Rank 4: s480128_mpnn13 (82 aa, 11/18 hotspots)

Hotspot	Binder Res	AA	Dist (Å)	Interaction Type
C:Y10	B:3, B:4, B:7	H, Q, P	2.6	Hydrophobic
C:Q15	B:8, B:11, B:12	K, A, W	2.2	**H-bond**, Hydrophobic
C:K16	B:12	W	3.5	Hydrophobic
E:E11	B:14	Q	4.9	vdW
E:Y10	B:3, B:7, B:10, B:11, B:14	H, P, W, A, Q	1.7	**H-bond**, Hydrophobic
E:Q15	B:11, B:14, B:15	A, Q, F	1.6	**H-bond**, Hydrophobic
E:K16	B:12, B:15	W, F	2.9	Hydrophobic
G:E11	B:14	Q	3.7	vdW
G:Y10	B:3, B:10, B:14	H, W, Q	1.9	**H-bond**, Hydrophobic
G:Q15	B:14, B:15, B:18	Q, F, F	1.9	Hydrophobic
G:K16	B:15, B:18	F, F	2.4	Hydrophobic

Key binder residues: B:3 H (H-bond with G:Y10), B:7 P (H-bond with E:Y10), B:11 A (H-bond with E:Q15), B:12 W (H-bond with C:Q15). Same scaffold as Rank 5. Zero Met at interface.

Sequence: SFHQKYPKAWAWQQFLEFIVRQILGDTPEAKKIVEEVTSEAEKLLEADKSGELGTTEEGANKLFIEMLTRAFSKVADVLLNP

3.5 Rank 5: s480128_mpnn17 (82 aa, 11/18 hotspots)

Hotspot	Binder Res	AA	Dist (Å)	Interaction Type
C:Y10	B:3, B:4, B:7	H, Q, P	2.4	Hydrophobic
C:Q15	B:11, B:12, B:57	A, W, E	2.1	**H-bond**, Hydrophobic
C:K16	B:12	W	3.3	Hydrophobic
E:E11	B:10	W	4.8	vdW
E:Y10	B:3, B:7, B:10, B:11	H, P, W, A	2.0	**H-bond**, Hydrophobic
E:Q15	B:11, B:12, B:15	A, W, F	2.2	**H-bond**, Hydrophobic
E:K16	B:12, B:15	W, F	2.8	Hydrophobic
G:E11	B:10	W	4.4	vdW
G:Y10	B:3, B:10, B:14	H, W, Q	1.9	**H-bond**, Hydrophobic
G:Q15	B:14, B:15, B:18	Q, F, F	2.0	Hydrophobic
G:K16	B:15, B:18, B:22	F, F, Q	2.3	Hydrophobic

Key binder residues: B:3 H (H-bond with G:Y10), B:7 P (H-bond with E:Y10), B:11 A (H-bond with E:Q15), B:57 E (H-bond with C:Q15). Lowest binder RMSD (0.90 Å). Zero Met at interface.

Sequence: SFHQKYPKAWAWIQFLRFIVEQILGDTPEAQDIYDTVASEAKEKLEADKSGELGTTEEGANELFIEMLTRAFSLVADVLLNP

3.6 Rank 6: s967366_mpnn11 (82 aa, 10/18 hotspots)

Hotspot	Binder Res	AA	Dist (Å)	Interaction Type
C:Q15	B:10, B:13	F, M	2.1	Hydrophobic
C:K16	B:10	F	2.9	Hydrophobic
E:Y10	B:13, B:17, B:20	M, D, Y	1.9	**H-bond**, Hydrophobic
E:Q15	B:10, B:13, B:14, B:17	F, M, F, D	2.1	Hydrophobic
E:K16	B:10, B:14	F, F	2.5	Hydrophobic
G:E11	B:21, B:25	A, R	2.4	Hydrophobic
G:H13	B:25	R	3.8	vdW
G:Q15	B:13, B:14, B:15, B:17, B:18	M, F, F, D, Y	2.2	**H-bond**, Hydrophobic
G:K16	B:14	F	2.6	Hydrophobic
G:Y10	B:20, B:21, B:24, B:25	Y, A, Y, R	2.3	Hydrophobic

Key binder residues: B:17 D (H-bond with E:Y10), B:14 F (H-bond with G:Q15). Highest pLDDT (0.94), best shape complementarity (0.76). Mostly hydrophobic contacts. Only design reaching G:H13 besides s716952.

Sequence: MPKEVEIWEFLQMFFMDYFYAEIYRGKLSEEEKEIVEKIDKTWQKVIDNMKKNNGVMSEEDQKEMQEVLLDIINLKKKLEEK

3.7 Rank 7: s311665_mpnn6 (68 aa, 9/18 hotspots — smallest)

Hotspot	Binder Res	AA	Dist (Å)	Interaction Type
C:Y10	B:3, B:6, B:7	R, L, T	2.6	Hydrophobic
C:Q15	B:6, B:7, B:12	L, T, M	2.6	Hydrophobic
C:K16	B:12	M	3.4	vdW
E:Y10	B:1, B:2, B:6	M, T, L	1.8	**H-bond**, Hydrophobic
E:Q15	B:11, B:12, B:15, B:16	F, M, D, M	2.4	Hydrophobic
E:K16	B:12, B:16	M, M	2.3	Hydrophobic
G:Y10	B:1, B:11, B:15, B:19, B:32, B:36	M, F, D, H, Y, I	2.0	**H-bond**, Hydrophobic
G:Q15	B:15, B:16, B:19, B:20	D, M, H, L	2.1	**H-bond**, Hydrophobic
G:K16	B:16, B:20	M, L	2.6	Hydrophobic

Key binder residues: B:15 D (H-bond with G:Y10 and G:Q15), B:1 M (H-bond with E:Y10). Smallest binder (68 aa, 7.5 kDa) — best candidate for bispecific fusion. Symmetric contacts (Y10+Q15+K16 on all 3 chains). Lowest surface hydrophobicity (0.30).

Sequence: MTREMLTDPWFMITDMIYHLFMKDNEEISKKYNEIIENADKMTPEEFREKLMELLVEAVRTWHKRNFE

4. Binder Interface Residues — Complete Lists

Binder residues within 5 Å of any target hotspot, classified by interaction type.

4.1 s120913_mpnn1 (24 interface residues contacting hotspots)

Res	AA	Interactions	Res	AA	Interactions
B:2	D	vdW	B:47	A	Hydrophobic
B:3	T	Hydrophobic	B:48	R	vdW
B:4	R	Hydrophobic	B:50	K	Hydrophobic
B:7	L	Hydrophobic	B:52	F	Hydrophobic
B:8	**W**	**H-bond**, Hydrophobic	B:74	H	vdW
B:9	W	vdW	B:75	K	vdW
B:10	F	vdW	B:76	I	Hydrophobic
B:11	**A**	**H-bond**	B:77	**V**	**H-bond**
B:12	**T**	**H-bond**, Hydrophobic	B:78	E	Hydrophobic
B:14	**Q**	**H-bond**, Hydrophobic	B:79	**E**	**H-bond, Salt bridge**
B:15	L	Hydrophobic			
B:18	**R**	**H-bond, Salt bridge**			

4.2 s120913_mpnn2 (24 interface residues contacting hotspots)

Res	AA	Interactions	Res	AA	Interactions
B:1	M	vdW	B:47	A	Hydrophobic
B:2	P	Hydrophobic	B:48	R	vdW
B:3	T	Hydrophobic	B:50	K	Hydrophobic
B:4	R	Hydrophobic	B:52	F	Hydrophobic
B:7	L	Hydrophobic	B:75	E	vdW
B:8	**W**	**H-bond**, Hydrophobic	B:76	**I**	**H-bond**, Hydrophobic
B:9	W	vdW	B:77	**V**	**H-bond**
B:10	F	vdW	B:78	E	Hydrophobic
B:11	**A**	**H-bond**, Hydrophobic	B:79	**E**	**H-bond**, Hydrophobic
B:12	**T**	**H-bond**, Hydrophobic			
B:14	**Q**	**H-bond**, Hydrophobic			
B:15	L	Hydrophobic			
B:18	**R**	**H-bond, Salt bridge**			

4.3 s716952_mpnn20 (19 interface residues contacting hotspots)

Res	AA	Interactions	Res	AA	Interactions
B:1	E	vdW	B:46	L	vdW
B:2	G	vdW	B:48	H	Hydrophobic

Res	AA	Interactions	Res	AA	Interactions
B:3	I	Hydrophobic	B:49	D	Hydrophobic
B:4	**P**	**H-bond**, Hydrophobic	B:50	D	Hydrophobic
B:5	P	vdW	B:51	S	vdW
B:6	**T**	**H-bond**, Hydrophobic	B:54	V	Hydrophobic
B:7	I	Hydrophobic	B:55	Y	Hydrophobic
B:8	V	Hydrophobic	B:58	M	Hydrophobic
B:9	E	vdW	B:59	L	Hydrophobic
B:43	H	vdW			

4.4 s967366_mpnn11 (10 interface residues contacting hotspots)

Res	AA	Interactions
B:10	F	Hydrophobic
B:13	M	Hydrophobic
B:14	**F**	**H-bond**, Hydrophobic
B:15	F	vdW
B:17	**D**	**H-bond**, Hydrophobic
B:18	Y	Hydrophobic
B:20	Y	vdW
B:21	A	Hydrophobic
B:24	Y	Hydrophobic
B:25	R	vdW

4.5 s311665_mpnn6 (13 interface residues contacting hotspots)

Res	AA	Interactions
B:1	**M**	**H-bond**, Hydrophobic
B:2	T	vdW
B:3	R	vdW
B:6	L	Hydrophobic
B:7	T	vdW
B:11	F	Hydrophobic
B:12	M	Hydrophobic
B:15	**D**	**H-bond**, Hydrophobic
B:16	M	Hydrophobic
B:19	H	Hydrophobic
B:20	L	Hydrophobic
B:32	Y	Hydrophobic
B:36	I	Hydrophobic

5. Design Comparison & Recommendations

5.1 Ranking by Application

Application	Best design	Why
Overall lead	s120913_mpnn1	Highest i_pTM (0.81), 14/18 hotspots, 4 salt bridges, clean
Strongest binding	s120913_mpnn2	dG=-98.9, 11.5 H-bonds, 3146 Å² BSA
Broadest chain G coverage	s716952_mpnn20	6/6 on chain G, only design contacting G:H13+H14
Highest confidence	s967366_mpnn11	pLDDT=0.94, B_pLDDT=0.95, B_RMSD=1.11
Bispecific fusion	s311665_mpnn6	Smallest (68 aa), lowest surf.hydro (0.30), clean
Structural consistency	s480128_mpnn17	B_RMSD=0.90 Å, most stable fold

5.2 Key Observations

- **4 unique scaffolds** produced 7 accepted designs (s120913: 2, s480128: 2, s967366: 1, s311665: 1, s716952: 1).
- **All designs are helical** (65–85% helix), with no significant beta-sheet content.
- **Salt bridges only in s120913 scaffold** — B:18 R with G:Q15 and B:79 E with E:K16. These electrostatic interactions likely contribute to its superior i_pTM.
- **Q15 and K16 are universal anchor points** — contacted by every design on every chain.
- **H13/H14 are druggability gaps** — only 2 of 7 designs contact them. Designs with these contacts (s120913, s716952) have the broadest hotspot coverage.
- **Binding is predominantly hydrophobic** — only 2/7 designs have salt bridges. H-bond counts range 4–13, while hydrophobic contact pairs range 142–203.

6. File Locations

File	Path
Accepted PDBs	`bindcraft/designs/Accepted/*.pdb`
Full metrics	`bindcraft/designs/final_design_stats.csv`
This report	`docs/accepted_models_info.pdf`
Design report (full campaign)	`docs/bindcraft_design_report.pdf`